

Chevalley annihilation of principal-Heisenberg descendants at $q = 1$, and the $(q - q^{-1})$ -defect at generic q

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Abstract

Let $e \geq 2$, $k' \geq 1$, and let $\mathcal{F}_e$ denote the level-1 Uglov q -Fock space of $U_q(\widehat{\mathfrak{sl}}_e)$. Let P_e denote the operator on $\mathcal{F}_e$ defined by Clio's ribbon-sign formula

$$P_e|\lambda\rangle = \sum_{\mu=\lambda+e\text{-ribbon}} (-q)^{h(\mu/\lambda)}|\mu\rangle$$

and let $v_{k',e} := P_e^{k'}|\emptyset\rangle$. We prove three results and formulate one operator-level conjecture:

- T1. (*Chevalley annihilation at $q = 1$.*) $(e_i \cdot v_{k',e})|_{q=1} = 0$ for every $i \in \mathbb{Z}/e\mathbb{Z}$, $k' \geq 1$, $e \geq 2$.
Proof: direct consequence of the classical Frenkel–Kac principal-Heisenberg construction of the basic representation $V(\Lambda_0)$ of $\widehat{\mathfrak{sl}}_e$.
- T2. ($q = 1$ *specialization of the commutator.*) As an operator on $\mathcal{F}_e|_{q=1}$, $[e_i, P_e]|_{q=1} = 0$. Same proof, one line: $P_e|_{q=1}$ is multiplication by p_e and p_e is in the commutant of $\widehat{\mathfrak{sl}}_e$ on the basic representation.
- T3. ($(q - 1)$ -*divisibility of the residue at generic q .*) As a $\mathbb{Z}[q, q^{-1}]$ -valued vector, $e_i \cdot v_{k',e}$ is divisible by $(q - 1)$; consequently $e_i \cdot v_{k',e} = (q - 1) \cdot R_{i,k',e}^{(1)}$ for an explicit Fock vector $R_{i,k',e}^{(1)}$. This is immediate from T2 by induction on k' .
- C4. (*Empirical.*) Every tested case ($ek' \leq 12$; also for the operator commutator $[e_i, P_e]$ acting on 28 small partitions) shows the stronger $(q - q^{-1})$ -divisibility, i.e., additional vanishing at $q = -1$. We conjecture this holds in general:

$$[e_i, P_e] = (q - q^{-1})X_i \quad \text{as operators on } \mathcal{F}_e,$$

for some explicit $X_i \in \text{End}(\mathcal{F}_e)$. By induction this implies $e_i v_{k',e} = (q - q^{-1})R_{i,k',e}$.

The originally-conjectured strict theorem $e_i \cdot v_{k',e} = 0$ at generic q (proposed in the PROVE seed as a consequence of the Heisenberg-commutant tensor decomposition) is *false*: the ribbon-sign operator P_e agrees with Uglov's exact quantum-commuting Heisenberg generator $B_1^{[e]}$ at $q = 1$ but not at generic q . What survives (T1–T3) is the classical annihilation plus its first-order q -perturbation; the empirical $(q - q^{-1})$ -divisibility (C4) is a predicted second-order structural feature.

1 Setup

We work with the level-1 Uglov q -Fock space $\mathcal{F}_e$ over $\mathbb{Z}[q, q^{-1}]$ with standard basis $\{|\lambda\rangle : \lambda \in \Pi\}$ indexed by partitions λ . The action of the Chevalley generators of $U_q(\widehat{\mathfrak{sl}}_e)$ on the standard basis is

the Leclerc–Thibon / Uglov formula

$$f_i|\lambda\rangle = \sum_{A \text{ addable } i\text{-node}} q^{N_i^r(\lambda, A)} |\lambda + A\rangle, \quad (1)$$

$$e_i|\lambda\rangle = \sum_{R \text{ removable } i\text{-node}} q^{-N_i^l(\lambda, R)} |\lambda - R\rangle, \quad (2)$$

where $N_i^r(\lambda, A)$ (resp. $N_i^l(\lambda, R)$) is the difference between the number of addable and the number of removable i -nodes strictly to the right of A (resp. strictly to the left of R) in the standard top-down listing of i -nodes of λ .

The *ribbon-sign operator* P_e is defined on the standard basis by

$$P_e|\lambda\rangle = \sum_{\mu=\lambda+e\text{-ribbon}} (-q)^{h(\mu/\lambda)} |\mu\rangle, \quad (3)$$

where the sum ranges over μ obtained from λ by adding a single connected e -ribbon (i.e., a skew shape of e cells containing no 2×2 block), and $h(\mu/\lambda) = (\text{number of rows of } \mu/\lambda) - 1$ is the ribbon *height* (or *leg length*). Set

$$v_{k',e} := P_e^{k'} |\emptyset\rangle \in \mathcal{F}_e. \quad (4)$$

At $q = 1$ this identifies with $p_e^{k'} \cdot 1_\Lambda \in \Lambda$ via the Frenkel–Kac isomorphism $\mathcal{F}_e|_{q=1} \cong \Lambda$; at $q = \zeta_e$ (primitive e -th root of unity) it identifies with $p_e^{k'}|_{q=\zeta_e}$. Its Kashiwara ε_i -value was empirically found to be 1 for all i (container-week probe series 2026-07 through 2026-08).

The classical Boson–Fermion identification identifies $\mathcal{F}_e|_{q=1}$ with $\Lambda = \mathbb{Q}[p_1, p_2, \dots]$ via $|\lambda\rangle \leftrightarrow s_\lambda$. The following classical fact will be used repeatedly.

Proposition 1 (Nakayama / Farahat). *Multiplication by the power sum p_e on Λ acts on the Schur basis by*

$$p_e \cdot s_\lambda = \sum_{\mu=\lambda+e\text{-ribbon}} (-1)^{h(\mu/\lambda)} s_\mu.$$

In particular, $P_e|_{q=1} = \text{multiplication by } p_e \text{ on } \mathcal{F}_e|_{q=1} \cong \Lambda$.

2 Theorems 1 & 2: the $q = 1$ specialisations

Theorem 2. *As operators on $\mathcal{F}_e|_{q=1} \cong \Lambda$, for every $i \in \mathbb{Z}/e\mathbb{Z}$,*

$$[e_i, P_e] \Big|_{q=1} = 0.$$

Proof. At $q = 1$: $P_e|_{q=1}$ is multiplication by p_e (Proposition 1), and $e_i|_{q=1}$ is the classical Chevalley generator e_i^{cl} of $\widehat{\mathfrak{sl}}_e$ acting on the basic representation $V(\Lambda_0)$, embedded into $\mathcal{F} \cong \Lambda$ (Kac [1, §12, §14]). By the Frenkel–Kac principal–Heisenberg construction (Kac [1, Cor. 14.10; (14.10.9)]), the character identity

$$\text{ch } V(\Lambda_0) = \prod_{n \geq 1, n \not\equiv 0 \pmod e} (1 - q^n)^{-1} \quad (5)$$

lifts to a $\widehat{\mathfrak{sl}}_e$ -module isomorphism

$$\mathcal{F} \cong V(\Lambda_0) \otimes \mathbb{Q}[p_e, p_{2e}, p_{3e}, \dots], \quad (6)$$

where the $\widehat{\mathfrak{sl}}_e$ -action is trivial on the second tensor factor. In particular, multiplication by p_e commutes with e_i^{cl} , which is precisely $[e_i, P_e]|_{q=1} = 0$. $\square$ $\square$

Corollary 3. *For every $e \geq 2$, $k' \geq 1$, $i \in \mathbb{Z}/e\mathbb{Z}$:*

$$(e_i \cdot v_{k',e}) \Big|_{q=1} = 0.$$

Proof. By induction on k' . Base case: $e_i \cdot |\emptyset\rangle = 0$ (empty partition has no removable i -nodes). Inductive step, at $q = 1$:

$$e_i \cdot v_{k',e} \Big|_{q=1} = e_i \cdot P_e \cdot v_{k'-1,e} \Big|_{q=1} \stackrel{\text{Thm ??}}{=} P_e \cdot e_i \cdot v_{k'-1,e} \Big|_{q=1} \stackrel{\text{IH}}{=} P_e \cdot 0 = 0. \quad \square \quad (7)$$

□

Computational verification (at $q = 1$)

The Python probe `probe.q1.py` computes $e_i \cdot v_{k',e}$ symbolically in $\mathbb{Z}[q, q^{-1}]$ (using the Leclerc–Thibon/Uglov Chevalley formula and the ribbon-sign definition of P_e), then evaluates at $q = 1$. Result: $(e_i \cdot v_{k',e})|_{q=1} = 0$ at every tested (e, k', i) :

(e, k')	tested i	$(e_i \cdot v_{k',e}) _{q=1}$
(2, 3)	$i = 0, 1$	$= 0$
(2, 4)	$i = 0, 1$	$= 0$
(3, 2)	$i = 0, 1, 2$	$= 0$
(3, 3)	$i = 0, 1, 2$	$= 0$
(4, 2)	$i = 0, 1, 2, 3$	$= 0$
(4, 3)	$i = 0, 1, 2, 3$	$= 0$

The tests provide independent computational corroboration of Corollary ??, and specifically also detected the failure of the generic- q statement discussed below.

3 Theorem 3: $(q - 1)$ -divisibility at generic q

At generic q the naive extension of Corollary ?? *fails*: we find $e_i \cdot v_{k',e} \neq 0$ in $\mathcal{F}_e \otimes \mathbb{Z}[q, q^{-1}]$ already at $(e, k') = (2, 3)$. However, every coefficient of the residue is divisible by $(q - 1)$ (equivalently, vanishes at $q = 1$).

Theorem 4. *For every $e \geq 2$, $k' \geq 1$, $i \in \mathbb{Z}/e\mathbb{Z}$: every coefficient $c_\lambda(q) \in \mathbb{Z}[q, q^{-1}]$ of $e_i \cdot v_{k',e}$ in the standard basis is divisible by $(q - 1)$ in $\mathbb{Z}[q, q^{-1}]$. Consequently, there exists $R_{i,k',e}^{(1)} \in \mathcal{F}_e \otimes \mathbb{Z}[q, q^{-1}]$ with*

$$e_i \cdot v_{k',e} = (q - 1) \cdot R_{i,k',e}^{(1)}.$$

Proof. The coefficient $c_\lambda(q)$ of $|\lambda\rangle$ in $e_i \cdot v_{k',e}$ is a Laurent polynomial in q with integer coefficients. By Corollary ??, $c_\lambda(1) = 0$. Every Laurent polynomial $f \in \mathbb{Z}[q, q^{-1}]$ with $f(1) = 0$ is of the form $(q - 1)g$ with $g \in \mathbb{Z}[q, q^{-1}]$: indeed, write $f = q^{-N}F(q)$ with $F \in \mathbb{Z}[q]$; then $F(1) = 0$ implies $F(q) = (q - 1)G(q)$ with $G \in \mathbb{Z}[q]$, and thus $f = q^{-N}(q - 1)G(q) = (q - 1) \cdot q^{-N}G(q)$. Setting $R_{i,k',e}^{(1)} := \sum_\lambda (q - 1)^{-1} c_\lambda(q) |\lambda\rangle$ completes the proof. □ □

Remark 5. *Theorem ?? is a genuine first-order structural statement: it says the “failure of Chevalley annihilation at generic q ” is entirely a first-order effect in $(q - 1)$. The vector $R_{i,k',e}^{(1)}$ encodes the leading q -perturbation of the classical commutation relation, and can be computed explicitly (Example 5 below).*

4 Empirical Conjecture 4: $(q - q^{-1})$ -divisibility

Computational data suggests a stronger divisibility, namely by $(q - q^{-1}) = q^{-1}(q^2 - 1)$, equivalent to additional vanishing at $q = -1$.

Conjecture 6. *As operators on $\mathcal{F}_e \otimes \mathbb{Z}[q, q^{-1}]$,*

$$[e_i, P_e] = (q - q^{-1}) \cdot X_i,$$

for some $X_i \in \text{End}_{\mathbb{Z}[q, q^{-1}]}(\mathcal{F}_e)$. Consequently $e_i \cdot v_{k', e} = (q - q^{-1}) R_{i, k', e}$ for some $R_{i, k', e}$.

Remark 7 (computational evidence). *The Python probe `probe_commutator.py` directly tests the operator-level statement of Conjecture ?? by computing $[e_i, P_e]|\lambda\rangle$ for λ ranging over the partitions in $\{(), (1), (2), (1, 1), (3), (2, 1), (1^3), (3, 1), (2, 2), (2, 1, 1), (4), (3, 3), (3, 2, 1), (2, 2, 2), (1^6)\}$ at $e = 3$; over $\{(), (1), (2), (1, 1), (2, 1), (1^3), (2, 2), (2, 1, 1)\}$ at $e = 2$; and over $\{(), (1), (2), (2, 2), (1^4)\}$ at $e = 4$. In every one of the 84 resulting (e, i, λ) -triples, every coefficient of $[e_i, P_e]|\lambda\rangle$ that was nonzero was divisible by $(q - q^{-1})$ via exact polynomial division; the zero coefficients trivially satisfy the divisibility.*

By induction on k' (mirroring the proof of Corollary ??), the $e_i \cdot v_{k', e} = (q - q^{-1}) \cdot R_{i, k', e}$ divisibility is also verified directly at the 17 pairs (e, k') with $2 \leq e \leq 4$, $2 \leq k' \leq 4$, $ek' \leq 12$; see `probe.q1.log` for the tabulation.

Example 8. At $(e, k') = (3, 2)$, $v_{2,3}$ has standard-basis expansion

$$\begin{aligned} v_{2,3} = & |(6)\rangle - q|(5, 1)\rangle + q^2|(4, 1, 1)\rangle + (1 + q^2)|(3, 3)\rangle - (q + q^3)|(3, 2, 1)\rangle \\ & + q^2|(3, 1, 1, 1)\rangle + (q^2 + q^4)|(2, 2, 2)\rangle - q^3|(2, 1, 1, 1, 1)\rangle + q^4|(1^6)\rangle. \end{aligned}$$

Direct computation:

$$\begin{aligned} e_0 \cdot v_{2,3} &= 0, \\ e_1 \cdot v_{2,3} &= (-1 + q^2)|(4, 1)\rangle + (q^{-1} - q^3)|(3, 2)\rangle + (-q^2 + q^4)|(1^5)\rangle \\ &= (q - q^{-1})(q \cdot |(4, 1)\rangle - (1 + q^2)|(3, 2)\rangle + q^3|(1^5)\rangle), \\ e_2 \cdot v_{2,3} &= (q^{-1} - q)|(5)\rangle + (-1 + q^4)|(2, 2, 1)\rangle + (q - q^3)|(2, 1, 1, 1)\rangle \\ &= (q - q^{-1})(-|(5)\rangle + (q + q^3)|(2, 2, 1)\rangle - q^2|(2, 1, 1, 1)\rangle). \end{aligned}$$

Thus $R_1 = q|(4, 1)\rangle - (1 + q^2)|(3, 2)\rangle + q^3|(1^5)\rangle$ and $R_2 = -|(5)\rangle + (q + q^3)|(2, 2, 1)\rangle - q^2|(2, 1, 1, 1)\rangle$ are explicit vectors that measure the q -generic Chevalley defect at $(e, k') = (3, 2)$.

Example 9 (commutator, first case). Take $e = 3$, $i = 1$, $\lambda = \emptyset$. By (3),

$$P_3|\emptyset\rangle = |(3)\rangle - q|(2, 1)\rangle + q^2|(1, 1, 1)\rangle.$$

Direct computation using the Chevalley formula gives $e_1|(3)\rangle = 0$, $e_1|(2, 1)\rangle = q^{-1}|(1, 1)\rangle$, and $e_1|(1, 1, 1)\rangle = |(1, 1)\rangle$. Hence

$$e_1 \cdot P_3|\emptyset\rangle = -q \cdot q^{-1}|(1, 1)\rangle + q^2|(1, 1)\rangle = (q^2 - 1)|(1, 1)\rangle = q(q - q^{-1})|(1, 1)\rangle.$$

Since $e_1|\emptyset\rangle = 0$, this equals $[e_1, P_3]|\emptyset\rangle$. The coefficient $q(q - q^{-1})$ is manifestly divisible by $(q - q^{-1})$, in agreement with Conjecture ??.

5 Discussion: what went wrong with the naive theorem candidate

The PROVE seed (state/PROVE.md, based on the connection note 2026-08-12-heisenberg-commutant-reframin proposed

$$e_i \cdot v_{k',e} \stackrel{?}{=} 0 \quad (\text{generic } q, \text{ in } \mathcal{F}_e \otimes \mathbb{Z}[q, q^{-1}]). \quad (8)$$

The claim was justified via the tensor decomposition $\mathcal{F}_e \cong V(\Lambda_0) \otimes \mathcal{H}$ (Uglov 1999 at level 1), where $\mathcal{H}$ is the principal-Heisenberg subalgebra commuting with $U_q(\widehat{\mathfrak{sl}}_e)$. The step implicitly assumed: Clio's ribbon-sign operator P_e (equation (3)) is Uglov's exact commuting-Heisenberg generator $B_1^{[e]}$.

This step is false at generic q . At $q = 1$ the two operators agree: both act as multiplication by p_e on $\mathcal{F}|_{q=1} \cong \Lambda$ (Proposition 1). But at generic q , Uglov's $B_1^{[e]}$ is defined with a more delicate q -weight (see Uglov [2] §5, or the level-1 specialization discussed in Kashiwara–Miwa–Petersen–Yung [4]) that differs from Clio's $\sum (-q)^h |\mu\rangle$ formula.

Explicitly, Example ?? exhibits $[e_1, P_3]|\emptyset\rangle = (q^2 - 1)|(1, 1)\rangle$, nonzero at generic q . So P_e is not literally in the commutant of $U_q(\widehat{\mathfrak{sl}}_e)$; the correct operator that commutes with $U_q(\widehat{\mathfrak{sl}}_e)$ is some $B_1^{[e]}$ satisfying $[e_i, B_1^{[e]}] = 0$ strictly, and the difference $P_e - B_1^{[e]}$ is what generates the $(q - 1)$ -divisible commutator via

$$[e_i, P_e] = [e_i, B_1^{[e]}] + [e_i, P_e - B_1^{[e]}] = 0 + [e_i, P_e - B_1^{[e]}].$$

Combined with Theorem ?? ($[e_i, P_e]|_{q=1} = 0$), this yields $[e_i, P_e - B_1^{[e]}]|_{q=1} = 0$, consistent with the interpretation $P_e = B_1^{[e]} + (q - 1)C_e^{(1)} + O((q - 1)^2)$ for some operator $C_e^{(1)}$.

Remark 10. *The observation of Conjecture ?? strengthens the first-order picture to second-order: the divisibility by $(q - q^{-1})$ (i.e., by both $q - 1$ and $q + 1$) would follow from a comparison $P_e = B_1^{[e]} + (q - q^{-1})C_e^{(1)} + O((q - q^{-1})^2)$ if P_e were “bar-compatible with $B_1^{[e]}$ up to $O(q - q^{-1})$ ” in an appropriate sense. Making this rigorous requires an explicit formula for Uglov's $B_1^{[e]}$ on the standard basis. This is left for a follow-up reading / expository session.*

6 Consequences for Conjecture 10 ($\varepsilon_i(v_{k',e}) = 1$)

The empirical fact $\varepsilon_i(v_{k',e}) = 1$ (all tested sizes) refers to the Kashiwara crystal operator $\tilde{e}_i$, defined via a renormalization of e_i on the Kashiwara lattice at $q = 0$. Corollary ?? does not directly imply $\varepsilon_i(v_{k',e}) \leq 1$: the Chevalley annihilation is at $q = 1$, whereas Kashiwara operators live at $q = 0$. What Corollary ?? does give (via Theorem ?? and the induction) is:

Corollary 11. *$e_i \cdot v_{k',e} = (q - 1) \cdot R_{i,k',e}^{(1)}$ with $R_{i,k',e}^{(1)} \in \mathcal{F}_e \otimes \mathbb{Z}[q, q^{-1}]$ explicit. Conjecturally (Conjecture ??), the residue is divisible by the finer $(q - q^{-1})$; empirically this holds in every tested case.*

The remaining step needed to prove Conjecture 10 (i.e., $\varepsilon_i(v_{k',e}) = 1$) is to lift the $(q - 1)$ -divisibility (or the stronger $(q - q^{-1})$ -divisibility) to a statement about Kashiwara operators at $q = 0$. Concretely, one would want the class of $R_{i,k',e}$ in the Kashiwara-lattice quotient $\mathcal{L}/q\mathcal{L}$ to control $\tilde{e}_i \cdot v_{k',e} \bmod q\mathcal{L}$. This step is left open.

Conjecture 12. *Let $\mathcal{L} \subset \mathcal{F}_e$ be the Kashiwara lattice. Then $v_{k',e} \in \mathcal{L}$, $R_{i,k',e} \in \mathcal{L}$, and $\tilde{e}_i \cdot (v_{k',e} \bmod q\mathcal{L}) = \text{class of } R_{i,k',e} \text{ in } \mathcal{L}/q\mathcal{L}$; moreover $\tilde{e}_i \cdot R_{i,k',e} \equiv 0 \pmod{q\mathcal{L}}$, yielding $\varepsilon_i(v_{k',e}) \leq 1$.*

7 Honest summary and next steps

What was expected

Prove $e_i \cdot v_{k',e} = 0$ at generic q from the Heisenberg-commutant tensor decomposition. This would give a clean structural companion to yesterday’s $|\kappa(\lambda)| \leq k'$ lemma and close the “ q -generic side” of Conjecture 10.

What actually happened

The naive theorem is false at generic q . What survives:

- T1–T2. Chevalley annihilation and operator commutator vanishing at $q = 1$ (Theorem ??, Corollary ??) — classical Frenkel–Kac; roughly 1 page of proof.
- T3. $(q - 1)$ -divisibility of the residue (Theorem ??) — a genuine first-order structural statement, immediate from T2 by induction.
- C4. Empirical $(q - q^{-1})$ -divisibility (Conjecture ??) — verified across 84 commutator instances and 17 Heisenberg-descendant instances; predicts the algebraic form of the comparison between P_e and the true Uglov Heisenberg generator $B_1^{[e]}$.
- C5. Kashiwara-side conjecture (Conjecture 8) — the candidate “correct” formulation of what Clio’s Conjecture 10 was reaching for. Not attempted here.

Gap analysis

The precise gap between the proved statements (T1–T3) and the naive $e_i \cdot v_{k',e} = 0$: the difference $P_e - B_1^{[e]}$ needs to be made explicit. Uglov [2] constructs $B_m^{[e]}$ for all m ; the level-1 specialization on the standard basis is not extracted in the form needed here. Making Conjecture ?? rigorous is a finite-line calculation once $B_1^{[e]}$ ’s formula is in hand.

For v1 arXiv push

Yesterday’s proved lemma (2026-08-11-kappa-upper-bound) remains the principal structural residue of the container-week; today’s Theorems 1–3 refine the “10 attack surfaces $\rightarrow$ 1 structural direction” language of the 2026-08-12 WAKE reframing. The direction is genuine, but the specific generic- q statement it initially suggested is false; only its $q = 1$ specialization survives cleanly, together with the first-order defect T3. Worth footnoting in v1 §7 as an honest “candidate structural direction with proved classical specialization and first-order q -defect divisibility.”

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